



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 7 • STANDARD 6.NOS.7

Reflect Points Across Axes

Lesson 7-9 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can reflect points across the x-axis and y-axis on the coordinate plane.

Language Objective: I can explain my work using the words reflection, x-axis, y-axis, and symmetry.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Reflection

x-axis

y-axis

Symmetry

Integer

Perpendicular



Tap any word to see its meaning and a picture.

1

A flip of a point or shape over an axis to make a mirror image is a

2

The horizontal number line on the coordinate plane is the

3

The vertical number line on the coordinate plane is the

4

When two halves are mirror images of each other, the figure has

5

A whole number or its opposite, with no fraction part, is an

6

Two lines that meet at a 90° square corner are .

Write About the Math

How do reflections help the game designer create a balanced, symmetric level?

¿Cómo ayudan los reflejos a la diseñadora del juego a crear un nivel equilibrado y simétrico?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Reflection**
Reflexión

☐ **x-axis**
Eje x

☐ **y-axis**
Eje y

☐ **Symmetry**
Simetría

☐ **Integer**
Número entero

Model: *Reflecting over the y-axis changes the ____ coordinate because ____.* — Student states reflecting over the y-axis flips the sign of the x-coordinate while the y-coordinate stays the same, so (3,2) becomes (-3,2).

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The enemy reflected over the y-axis is at ____ because ____.** **The enemy reflected over the x-axis is at ____ because ____.** **Reflections help game designers because ____.**
- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **My evidence is ____, so ____.**
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Reflection
2. **Notes 2:** x-axis
3. **Notes 3:** y-axis
4. **Notes 4:** Symmetry
5. **Notes 5:** Integer
6. **Notes 6:** Perpendicular
7. **Watch for this mistake:** *A common mistake in Reflect Points Across Axes is flipping the sign of the wrong coordinate — changing the x-coordinate when reflecting over the x-axis, instead of the y-coordinate (or vice versa for the y-axis). For example, to reflect (3, -5) over the x-axis, a student might flip the x-coordinate and write (-3, -5), but reflecting over the x-axis only changes the sign of the y-coordinate: (3, -5) becomes (3, 5). Before you submit, ask: 'Did I flip the sign of the coordinate that matches the axis I'm reflecting over — y for the x-axis, x for the y-axis?'*
8. **Try It 1:** A. (5, 3) — *Reflecting across the y-axis flips the sign of the x-coordinate only. (-5, 3) becomes (5, 3).*
9. **Try It 2:** A. The y-coordinate — *Across the x-axis the rule is $(x, y) \rightarrow (x, -y)$, so only the y-coordinate flips sign. (8, -6) becomes (8, 6).*